

SIMATS ENGINEERING ASSESSMENT TOOL

COURSE NAME: Theory of Computation

COURSE CODE: CSA13 01

COURSE OUTCOME COVERED

CO1: Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems.(BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5
Duration : 1 Hour

Total Marks:25

Analytical Problem

Code of Conduct:

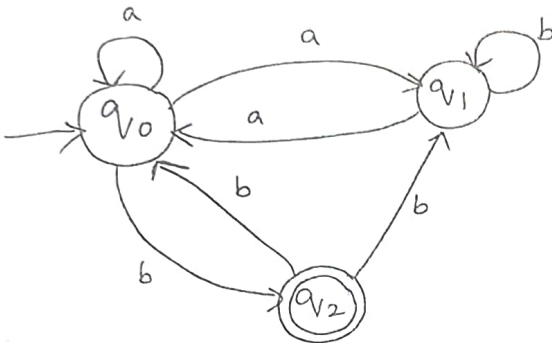
I Deepak V-192411021 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Deepak

SIMATS ENGINEERING ASSESSMENT TOOL

1. Consider the NFA given below which accepts some language L over the alphabet $\Sigma = \{a, b\}$. Construct a DFA that represents the same language L

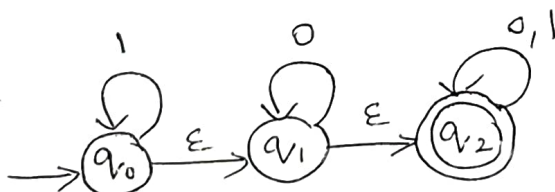


2. Let L be the language of words of length at least 2 whose last two letters are different from each other. For example $abaab$ is in L but not bb . Construct an NFA for L . Then draw the equivalent DFA

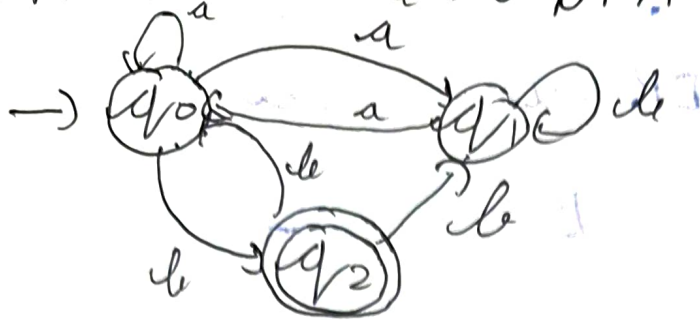
3. Design a Nondeterministic Finite Automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string **01011** is accepted by the NFA

4. Consider the NFA given below which accepts the language representing the set of all strings having 001 as a substring over the alphabet $\Sigma = \{0, 1\}$. Construct a DFA that represents the same language.

5. A NFA with ϵ -transitions is given below. Find ϵ -closure for each state and Design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states



1) NFA without ϵ to DFA



Sol:

$$\delta(q_0, a) = \delta(q_0, a)$$

$$\delta(q_0, a) = \{q_0, q_1\}$$

$$\delta(q_0, b) = \delta(q_0, b) \cup \delta(q_1, b)$$

$$\delta(q_0, b) = \{q_1, q_2\}$$

$$\delta(q_0, q_1, a) = \{q_0, q_1\}$$

$$\delta(q_0, q_2, b) = \{q_0, q_1, q_2\}$$

$$\delta(q_1, q_2, a) = \{q_0\}$$

$$\delta(q_1, q_2, b) = \{q_0, q_1, q_2\}$$

$$\delta(q_0, q_2, a) = \{q_0, q_1\}$$

$$\delta(q_0, q_2, b) = \{q_0, q_1, q_2\}$$

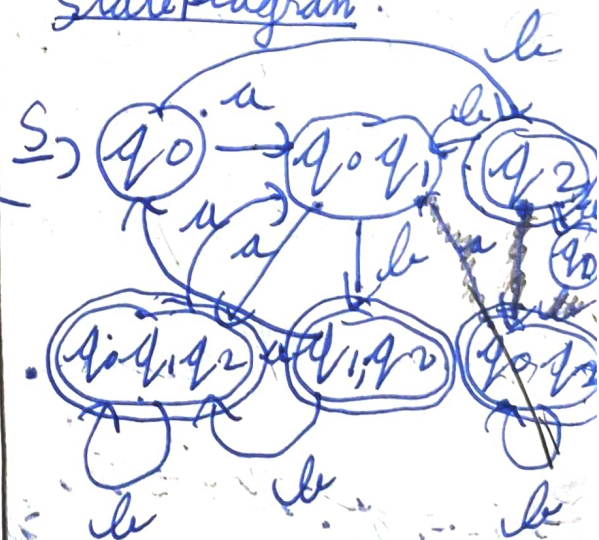
Transition Table:

State \ IP	a	b
q_0	$\{q_0, q_1\}$	$\{q_1, q_2\}$
q_1	$\{q_0\}$	$\{q_1, q_2\}$
q_2	$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$

DFA Transition Table:

State \ IP	a	b
$\{q_0\}$	$\{q_0, q_1\}$	$\{q_1, q_2\}$
$\{q_0, q_1\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$
$\{q_2\}$	ϕ	$\{q_0, q_1, q_2\}$
$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$
$\{q_1, q_2\}$	$\{q_0\}$	$\{q_0, q_1, q_2\}$
$\{q_0, q_2\}$	$\{q_0, q_1\}$	$\{q_0, q_1, q_2\}$

State Diagram:



$$\begin{aligned}
 f'(B, b) &\Rightarrow C_0 - \text{closure}(f(q_1, b) \cup \dots \cup f(q_{12}, b)) \\
 &\Rightarrow C_0 - \text{closure}(q_5 \cup q_{11} \cup q_{13}) \\
 &\Rightarrow (q_1, q_2, q_4, q_5, q_6, q_7, q_8, q_9, \\
 &\quad q_{11}, q_{12}, q_{13}, q_{15}) = D -
 \end{aligned}$$

$$\begin{aligned}
 f'(C, a) &\Rightarrow C_0 - \text{closure}(f(q_1, a) \cup \dots \cup f(q_{15}, a)) \\
 &\Rightarrow C_0 - \text{closure}(q_3 \cup q_{10} \cup q_{14}) \\
 &\Rightarrow (q_1, q_2, \overset{q_3}{q_4}, q_5, q_7, q_8, q_9, \\
 &\quad q_{10}, q_{12}, q_{14}, q_{15}) = E -
 \end{aligned}$$

$$\begin{aligned}
 f'(C, b) &\Rightarrow C_0 - \text{closure}(q_5 \cup q_{13}) \\
 &\Rightarrow C -
 \end{aligned}$$

$$\begin{aligned}
 f'(A, a) &\Rightarrow C_0 - \text{closure}(q_3 \cup q_{10} \cup q_{14}) \\
 &\Rightarrow E -
 \end{aligned}$$

$$\begin{aligned}
 f'(D, b) &\Rightarrow C_0 - \text{closure}(q_5 \cup q_{13}) \\
 &\Rightarrow C -
 \end{aligned}$$

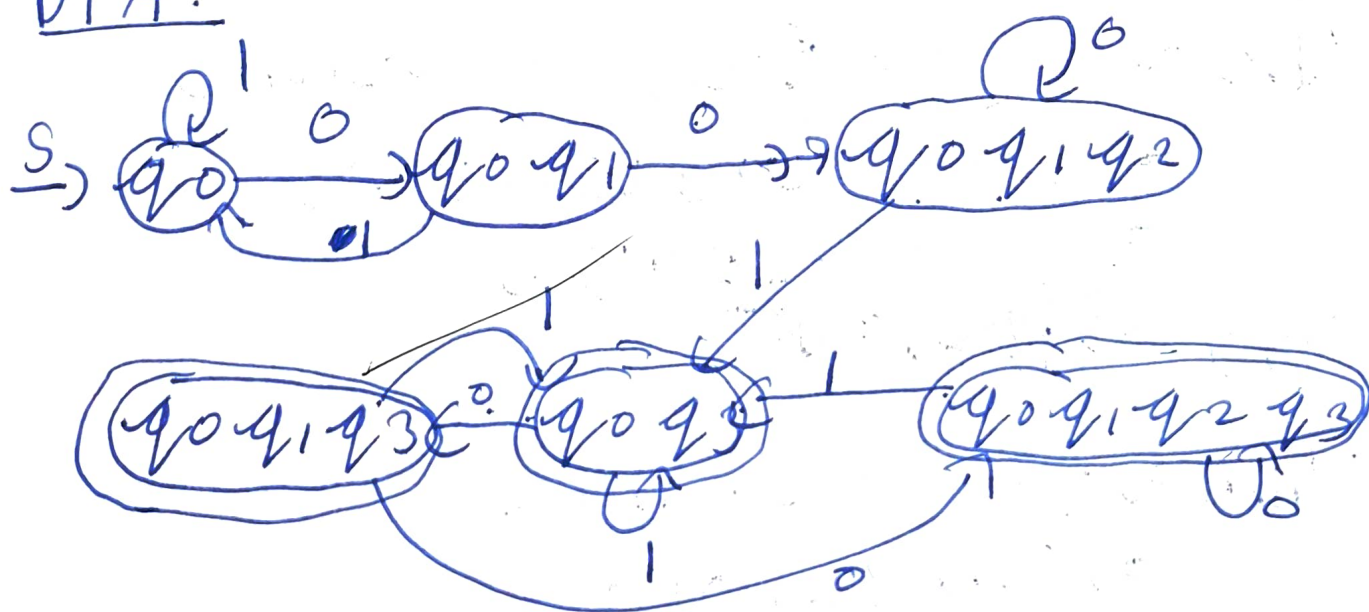
$$\begin{aligned}
 f'(E, a) &\Rightarrow C_0 - \text{closure}(q_3 \cup q_{10}) \\
 &\Rightarrow A -
 \end{aligned}$$

$$\begin{aligned}
 f'(E, b) &\Rightarrow C_0 - \text{closure}(q_5 \cup q_{11} \cup q_{13}) \\
 &= D -
 \end{aligned}$$

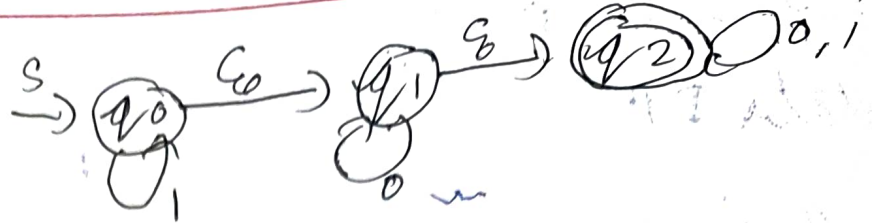
DFA Transition Table:

State & P	0	1
q_0	$\{q_0, q_1\}$	q_0
$\{q_0, q_1\}$	$\{q_0, q_1, q_2\}$	q_0
$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_3\}$
$\{q_0, q_3\}$	$\{q_0, q_1, q_3\}$	$\{q_0, q_3\}$
$\{q_0, q_1, q_3\}$	$\{q_0, q_1, q_2, q_3\}$	$\{q_0, q_3\}$
$\{q_0, q_1, q_2, q_3\}$	$\{q_0, q_1, q_2, q_3\}$	$\{q_0, q_3\}$

DFA:



5) NFA to DFA:



$$\epsilon\text{-closure}(q_0) \Rightarrow \{q_0, q_1, q_2\} = A -$$

$$\begin{aligned} \delta'(A, 0) &\Rightarrow \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(A), 0)) \\ &\Rightarrow \epsilon\text{-closure}(\delta(\{q_0, q_1, q_2\}, 0)) \\ &\Rightarrow \epsilon\text{-closure}(\delta(q_0, 0) \cup \delta(q_1, 0) \cup \delta(q_2, 0)) \\ &\Rightarrow \epsilon\text{-closure}(\{q_1, \phi \cup q_1 \cup q_2\}) \\ &\Rightarrow \epsilon\text{-closure}(\{q_1, q_2\}) \end{aligned}$$

$$\delta'(A, 0) \Rightarrow \{q_1, q_2\} = B -$$

$$\delta'(A, 1) \Rightarrow \epsilon\text{-closure}(\{q_0, q_2\})$$

$$\delta(A, 1) \Rightarrow \epsilon\text{-closure}(\{q_0, q_1, q_2\}) \Rightarrow A -$$

$$\delta'(B, 0) \Rightarrow \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(B), 0))$$

$$\Rightarrow \epsilon\text{-closure}(\delta(q_0, 0) \cup \delta(q_2, 0))$$

$$\Rightarrow \epsilon\text{-closure}(\phi \cup q_2)$$

$$\delta'(B, 0) \Rightarrow \{q_2\} \Rightarrow B^c$$

$$\delta'(B, 1) \Rightarrow \epsilon\text{-closure}(\{q_1, q_2\})$$

$$\Rightarrow \{q_1, q_2\} = B -$$

$$\delta'(C, 0) \Rightarrow \epsilon\text{-closure}(\{q_0, q_2\})$$

$$\Rightarrow \{q_0, q_1, q_2\} \Rightarrow B -$$

2) NFA with q_0 to DFA:

Forming Regular Expression: $(a+be)^* (ab+ba)$



$$\epsilon\text{-closure}(q_0) \Rightarrow \{q_0, q_1, q_2, q_4, q_6, q_8, q_{10}, q_{12}\} = A -$$

$$\begin{aligned} \delta'(A, a) &\Rightarrow \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(A), a)) \\ &\Rightarrow \epsilon\text{-closure}(\delta(q_0, a) \cup \delta(q_1, a) \dots \cup \delta(q_{12}, a)) \\ &\Rightarrow \epsilon\text{-closure}(\emptyset \cup \dots \cup q_3 \cup q_{10} \dots \cup \emptyset) \\ &\Rightarrow \{q_1, q_2, q_3, q_4, q_6, q_7, q_8, q_9, \\ &\quad q_{10}, q_{12}\} = B - \end{aligned}$$

$$\begin{aligned} \delta'(A, b) &\Rightarrow \epsilon\text{-closure}(\delta(q_0, b) \cup \delta(q_1, b) \dots \cup \delta(q_{12}, b)) \\ &\Rightarrow \epsilon\text{-closure}(\emptyset \cup q_5 \cup q_{13}) \end{aligned}$$

$$\begin{aligned} &\Rightarrow \{q_1, q_2, q_4, q_5, q_6, q_7, q_8, q_9, \\ &\quad q_{12}, q_{13}\} = C - \end{aligned}$$

$$\begin{aligned} \delta'(B, a) &\Rightarrow \epsilon\text{-closure}(\delta(q_{10}, a) \cup \delta(q_{12}, a) \dots \cup \delta(q_{14}, a)) \\ &\Rightarrow \epsilon\text{-closure}(q_3 \cup q_{10}) = B - \end{aligned}$$

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